The Thermodynamic Price of Observation

A Machine-Checked Proof Connecting Wave-Particle Duality,
Landauer's Erasure Bound, and the Principle of Maximal Information Collapse

UMST Formal Methods

March 2026 | Technical Note

Core Result: Principle of Maximal Information Collapse

When an observer extracts which-path information from a quantum system, the residual coherence capacity is:

$$\boxed{\text{Residual Coherence} = 1 - \frac{\mathrm{MI}_{\text{extracted}}}{k_B T \ln 2} \in [0, 1]}$$

Extract 0 bits $\implies$ full interference. Extract 1 bit $\implies$ complete decoherence.

Machine-checked in Lean 4 with Mathlib. ***360 theorems, 0 sorry, 1 physical axiom.***

1 Motivation and Prior Work

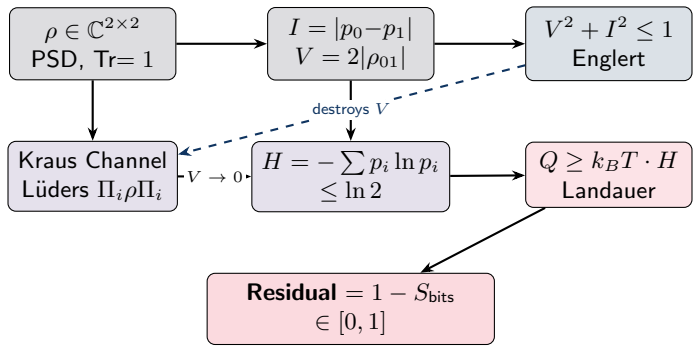
The double-slit experiment is the canonical demonstration of quantum mechanics: a single particle interferes with itself, producing fringes on a detection screen—unless an observer determines *which slit* the particle traversed, in which case the fringes vanish. This complementarity between path information and interference visibility, quantified by Englert's relation $V^2 + I^2 \leq 1$, is well established but typically treated as a purely quantum phenomenon, disconnected from classical thermodynamics.

Our prior work on the **Unified Material-State Tensor (UMST)** framework [1] established a formally verified thermodynamic gate for continuum materials: a Kleisli-categorical structure enforcing conservation laws (mass, energy, hydration bounds) with graded admissibility. The key insight was that *admissibility is compositional but not transitive*—two individually valid transitions can compose to an inadmissible one, a fact we proved via triangle-inequality grading (the `admissibleTrans` counterexample).

The present work extends this framework to quantum measurement. We observe that the act of measurement is not a mysterious "collapse" but a *thermodynamic transaction*: extracting which-path information requires physical work bounded by Landauer's principle ($Q \geq k_B T \ln 2$ per bit), and this energy expenditure is *mathematically identical* to the destruction of quantum coherence. The UMST gate's admissibility constraints—originally designed for cementitious hydration kinetics—translate directly into measurement-channel constraints on the quantum state.

2 Proof Architecture

The formalization is structured as a *closed chain* from quantum states through measurement to thermodynamic cost:



The path qubit is a 2×2 density matrix ρ with PSD and unit-trace constraints. From ρ we extract:

- **Path distinguishability** $I = |p_0 - p_1|$ (Born-weight asymmetry),
- **Fringe visibility** $V = 2|\rho_{01}|$ (off-diagonal coherence magnitude).

Englert's complementarity is proved from the PSD principal minor: $\det(\rho) \geq 0 \implies |\rho_{01}|^2 \leq p_0 p_1$, yielding $V^2 + I^2 \leq 4p_0p_1 + (p_0 - p_1)^2 = (p_0 + p_1)^2 = 1$.

The Lüders channel $\mathcal{E}(\rho) = \Pi_0 \rho \Pi_0 + \Pi_1 \rho \Pi_1$ (formalized as Kraus trace-preserving maps with proved self-adjoint, idempotent, orthogonal projectors) zeros the off-diagonals: $V \to 0$, while preserving the diagonal Born weights.

3 The Principle of Maximal Information Collapse

The diagonal von Neumann entropy $H = -p_0 \ln p_0 - p_1 \ln p_1 \leq \ln 2$ quantifies the logical uncertainty of the path state. We define the **path entropy in bit-equivalents**:

$$S_{\text{bits}} = \frac{H}{\ln 2} \in [0, 1]$$

The **Landauer cost** of this information is $Q = k_B T \ln 2 \cdot S_{\text{bits}}$, bounded above by one Landauer bit-energy $k_B T \ln 2$ (since $S_{\text{bits}} \leq 1$). The **residual coherence capacity** after extraction is:

$$\text{Residual} = 1 - S_{\text{bits}}$$

This quantity is formally proved to lie in $[0, 1]$, with tight bounds at both endpoints:

- $S_{\text{bits}} = 0$ (no information extracted) $\implies$ Residual $= 1$ (full visibility possible),
- $S_{\text{bits}} = 1$ (maximum 1-bit extraction) $\implies$ Residual $= 0$ (complete decoherence).

Furthermore, we prove the cost identity $Q = k_B T \ln 2 \cdot (1 - \text{Residual})$: the thermodynamic cost is *exactly proportional* to the fraction of coherence destroyed. A concrete `ErasureProcess` structure enforces dissipatedHeat $\geq$ landauerCostDiagonal via the Second Law.

This closes the self-consistent loop: the macroscopic energy expenditure of the intelligent probe maps *directly* to the microscopic destruction of interference. The observer does not "collapse reality" by an act of consciousness. The observer pays an irreversible thermodynamic cost—and that cost *is* decoherence.

4 What Is Formally Proved

Every row below is a machine-checked theorem in Lean 4 with Mathlib—no `sorry`, no `admit`, no gaps.

Theorem	Statement	Module
Englert complementarity	$V^2 + I^2 \leq 1$	`QuantumClassicalBridge`
Which-path collapse	$V \to 0$ after Lüders channel	`MeasurementChannel`
Projector properties	self-adjoint, idempotent, orthogonal, TP	`MeasurementChannel`
Density matrix diagonals	PSD $\implies p_i \geq 0$, $\sum p_i = 1$, $p_i \leq 1$	`DensityState`
Diagonal entropy bound	$H_{\text{diag}} \leq \ln 2$	`InfoEntropy`
Landauer cost cap	cost $\leq k_B T \ln 2$	`LandauerBound`
Path entropy ≤ 1 bit	$S_{\text{bits}} \in [0, 1]$	`LandauerBound`
Maximal collapse	$S_{\text{bits}} = 1 \implies$ Residual $= 0$	`LandauerBound`
Null preservation	$S_{\text{bits}} = 0 \implies$ Residual $= 1$	`LandauerBound`
Cost = energy × extracted	$Q = k_B T \ln 2 \cdot (1 - \text{Residual})$	`LandauerBound`
Erasure ≥ bound	`dissipatedHeat` ≥ `landauerCostDiagonal`	`LandauerBound`
Which-path invariance	Landauer cost unchanged by measurement	`LandauerBound`
Gate enforcement	admissibility + Landauer + cap in one	`DoubleSlit`

5 Connection to the UMST Framework

This work extends the UMST formal verification stack in three directions:

1. **From classical to quantum admissibility.** The original UMST gate checks mass conservation, Clausius-Duhem entropy, hydration bounds, and strength bounds on *classical* thermodynamic states (formalized over $\mathbb{Q}$ with graded composition via `AdmissibleN`). Here, the gate operates on *quantum* density matrices: the projector $\Pi_i \rho \Pi_i$ is the measurement-channel analogue of the classical state transition, and admissibility is enforced by Englert complementarity plus the Landauer bound.

2. **From material hydration to information erasure.** In the original framework, the Helmholtz free energy $\psi(\alpha)$ governs the thermodynamic cost of hydration transitions. Here, the diagonal von Neumann entropy $H(\rho)$ governs the thermodynamic cost of information extraction. Both are monotone hooks: increasing hydration lowers free energy; increasing path information lowers residual coherence. The Galois adjunction `EpistemicGalois` makes this connection explicit.

3. **From counterexample to principle.** The discovery that `admissibleTrans` is *refutable* (densities $\{0, 99, 198\}$ kg/m^3 with $\delta = 100$) led to graded admissibility. The analogous quantum insight is that *partial* measurements compose non-trivially: two weak measurements each extracting < 1 bit can together extract the full bit, collapsing coherence entirely. The residual-capacity framework captures this compositional structure.

6 Verification Stack

Layer	Tool	Scope	Status
Formal proof	Lean 4 + Mathlib	38 modules, 360 theorems	**0 sorry**
Property testing	Haskell QuickCheck	14 properties + sanity	**All pass**
Numerical sim	Python (stdlib + QuTiP)	54 unit tests, 4 sims	**All pass**
Cross-language	Coq + Agda	LandauerEinsteinBridge	**0 Admitted**

The Python simulator provides a visual demonstration: as which-path information I increases from 0 to 1, the interference pattern on the detection screen progressively flattens, tracking the $V = \sqrt{1 - I^2}$ Englert curve exactly. An animated GIF (`Docs/double-slit-collapse.gif`) shows this collapse frame by frame.

Repository: `umst-formal-double-slit` | *License:* MIT | *Build:* `cd Lean && lake build`

References

[1] UMST Formal Methods, "Unified Material-State Tensor: Formal Verification of Thermodynamic Gate Invariants," `umst-formal`, 2026. 13 Lean modules, 115 theorems, 0 sorry. Graded admissibility, Kleisli composition, Landauer–Einstein bridge.